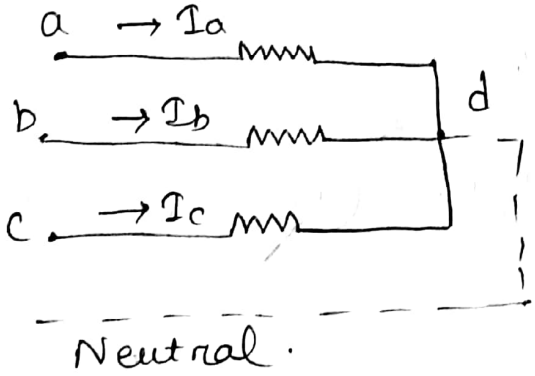
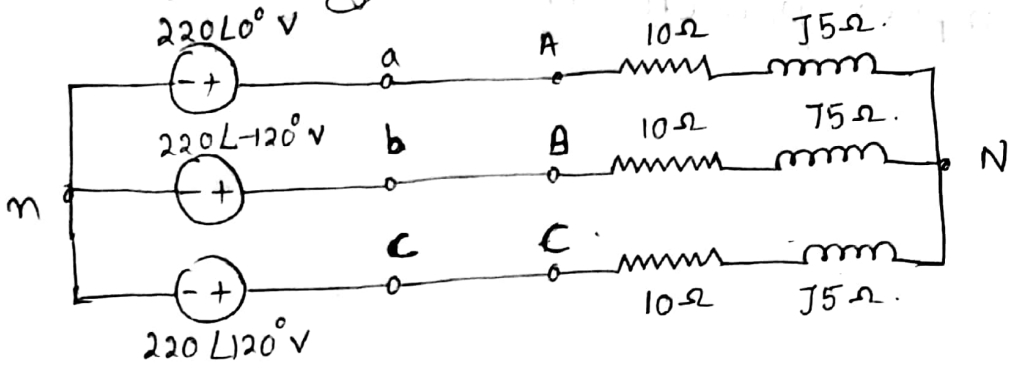
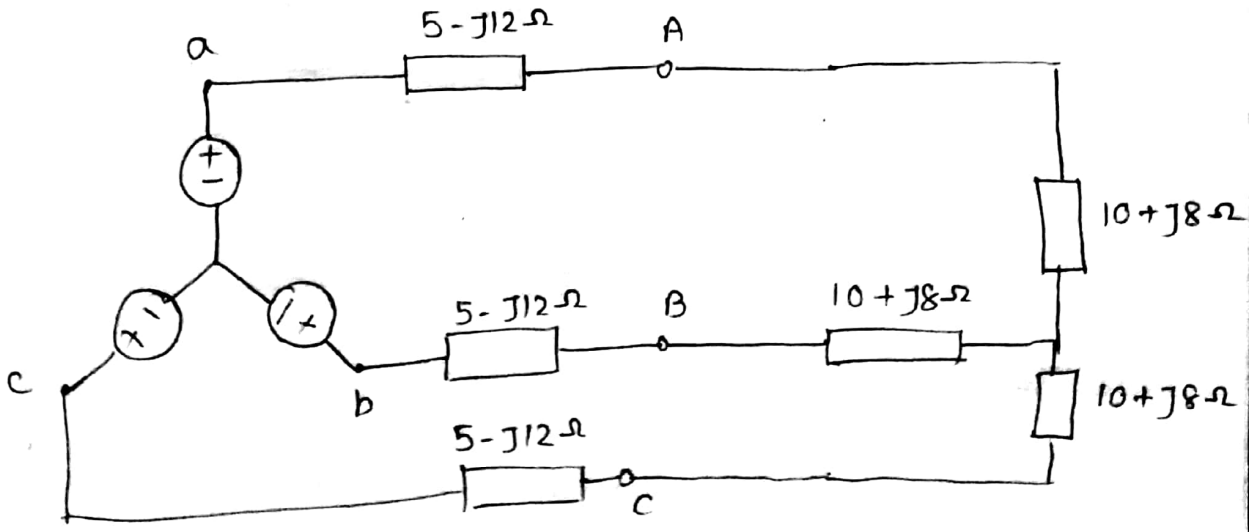
1) For the Y connected load, each of the three loads is a heater rated 5KW & the line voltage is 480 V AC. Find the line & phase currents & total power.



2) For the Y-Y circuit, find the line currents the line voltages & the load voltages.



3) Calculate the line currents in the 3 wire Y-Y system.



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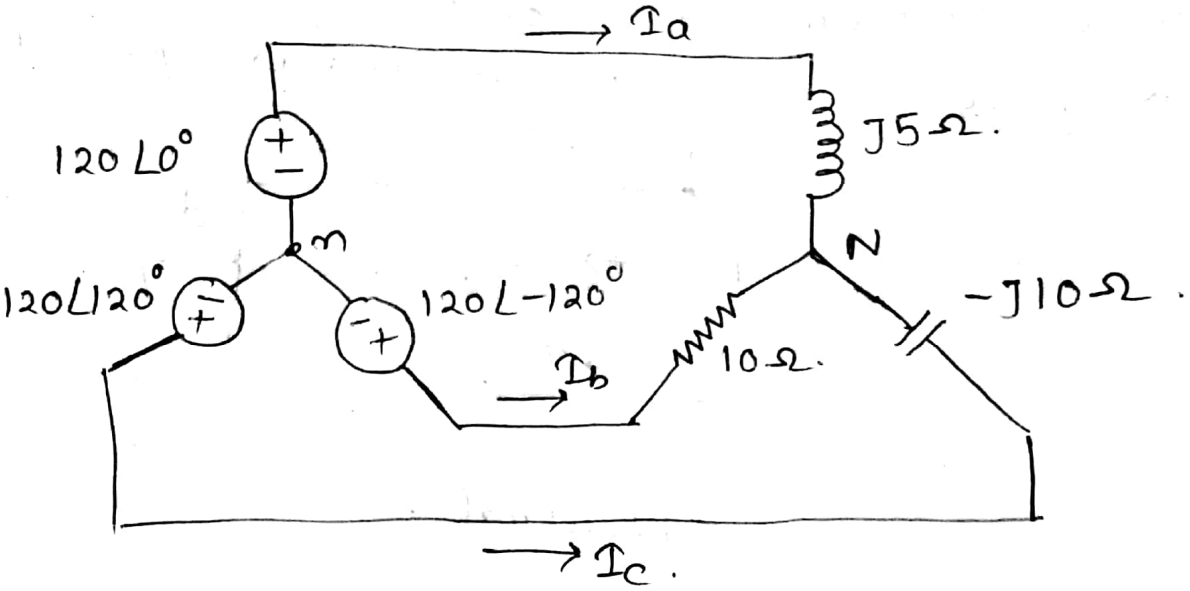
4) For the unbalanced circuit shown below, find the line currents.



5) In the above figure, find the total complex power absorbed by the load.

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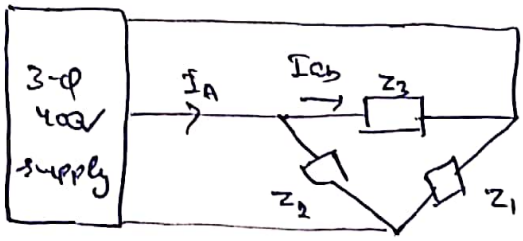
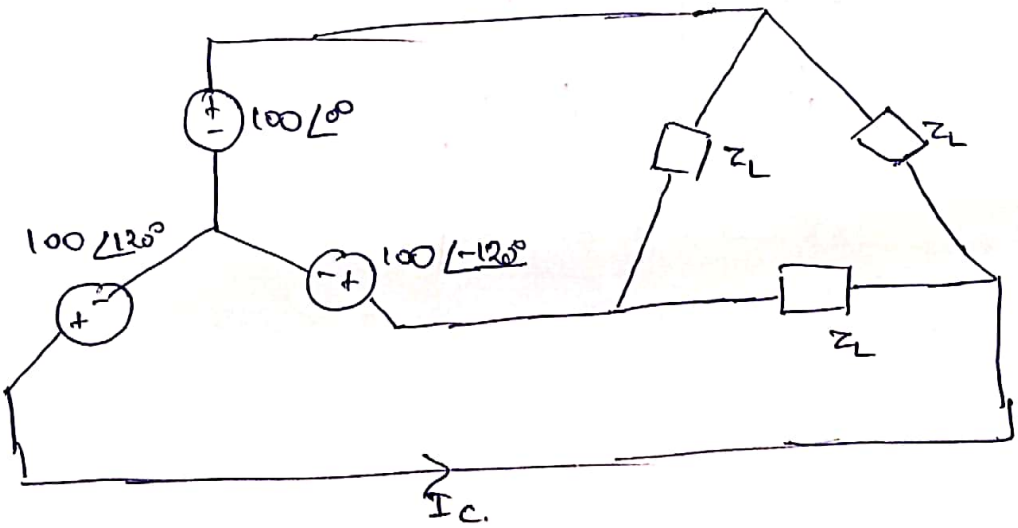
Assignment (4)

(6) A balanced three phase 415 V supply is connected to three phase delta connected load as shown in the figure. If $I_{ab} = 10\angle -30^\circ$ A. then what is the value of I_A ?



(7) Three identical inductors are connected in delta with a 240V, 60Hz, 3-phase supply. If the line current is $12\sqrt{3}$ A then what is the value of each of the inductance ?

(8) What is the value of <u>line current I_c</u> in wye - delta circuit shown in the figure where $Z_L = 30\angle 60^\circ\ \Omega$.



Q9 A set of three equal resistors each of value R_x connected in star across RYB of given figure consume the same power as the unbalanced delta connected load

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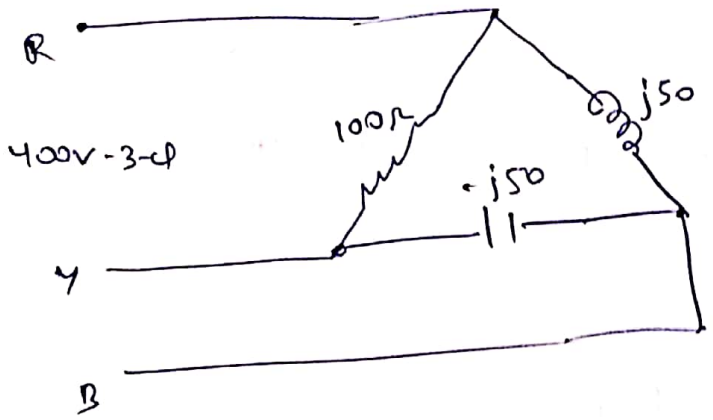
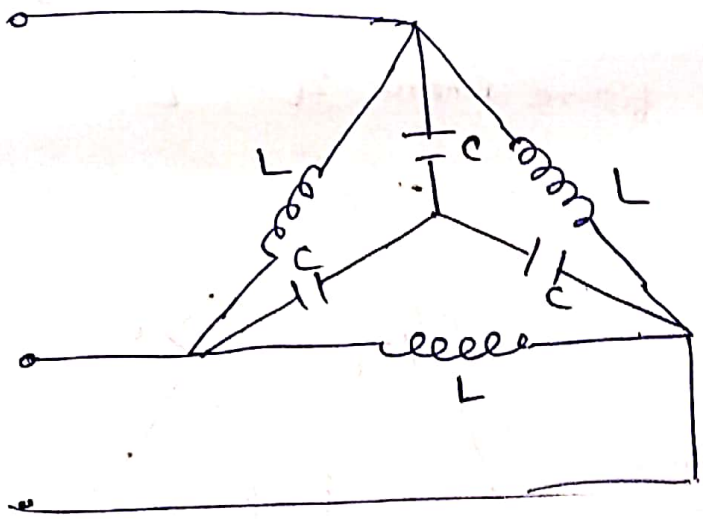
What is the value of R_x?



(Q10) In the balanced 3-phase 50Hz circuit shown below the value of the inductance is 10mH. The value of the capacitance for which all the line currents are zero in millifarads, is _____ ?



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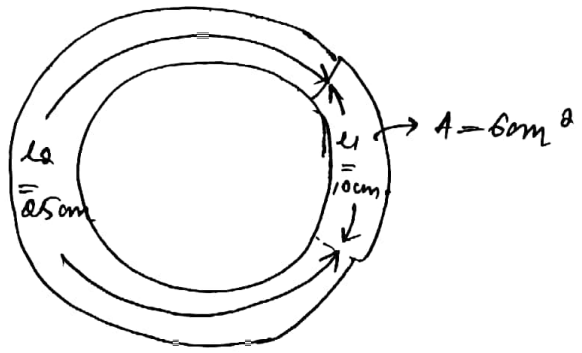
Q11. A circular iron ring has mean circumference 1.5m & cross sec area 0.01m². A saw cut of 4mm wide is made in ring. Calculate magnetising current required to produce flux of 0.8 mWb in air gap if ring is wound with a coil of 175 turns. ($\mu_{r\,iron} = 400$, leakage factor = 1.25) ?

Q12 A field coil has to develop 1500 AT with 60V applied. mean length of rectangular coil is 50cm. Calculate the wire size. Resistivity of copper = $2\times10^{-6}\,\Omega$cm.

Q13. A steel ring of 30cm mean diameter & circular section 2cm in diameter has air gap 1mm long. Its wound with 600 turns of wire carrying 2.5A. Find
(i) total mmf
(ii) Reluctance (total)
(iii) flux
The iron path takes 40% of total mmf.

Q14. The ring shaped core shown is made of material having ($\mu = 1000$). Flux density in thicker section is 1.5T. If current through the coil is not to exceed 0.5A, find no. of turns in the coil?



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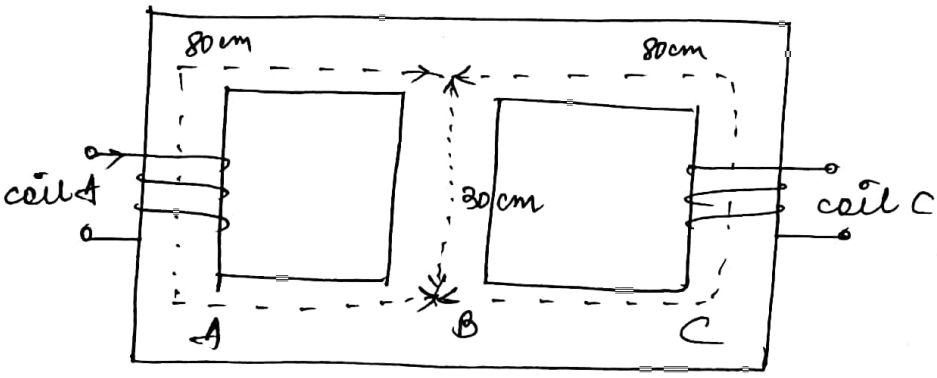
Q 15. A magnetic core made of sheet steel has dimensions as shown. The x-section is 25 cm² everywhere. Flux in branches A & B are 3500 μWb but in C is zero. Find the required AT for coils A & C. ($\mu_{steel} = 1000$)



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